

Technical Report

Gold and Oil Market Intelligence: evidence behind every claim

1. How this report is evidenced

Every figure and verdict below is backed by something that was actually run. The data pipeline from the SQL scripts was executed end to end in Snowflakes against the real source files, keeping the same structure (raw, clean, analytics, tests) and the same statistical functions, including the normal-approximation p-value macro defined in the framework script. The charts come from the analysis notebook, which was run on the same data. Where the pipeline produced no rows, that empty result is reported as it is, rather than smoothed over.

2. Pipeline execution and data quality

The pipeline ran without manual intervention and logged a clean quality report. The monthly fact table spans 2011-12 to 2018-12, 85 months. The outlier rule (a daily close more than three standard deviations from the mean) flagged nothing, which is expected: a single global z-score is a weak test on a series that trends across several years, so a zero count is a property of the rule rather than proof the data is perfect.

Dataset	Raw rows	Clean rows	Outliers ($ z >3$)
gold	1718	1718	0
oil	1844	1844	0
cpi	953	952	0
eia(world)	35	35	0

Table 1. QUALITY_REPORT, produced by the executed pipeline.

3. The coverage finding

Before any cross-asset test, the pipeline and the notebook both check whether the datasets share a time period. They do not. The price files end in December 2018 and the dollar and volatility files begin in June 2021, with no overlapping month. This is why several hypothesis tests below return no rows.

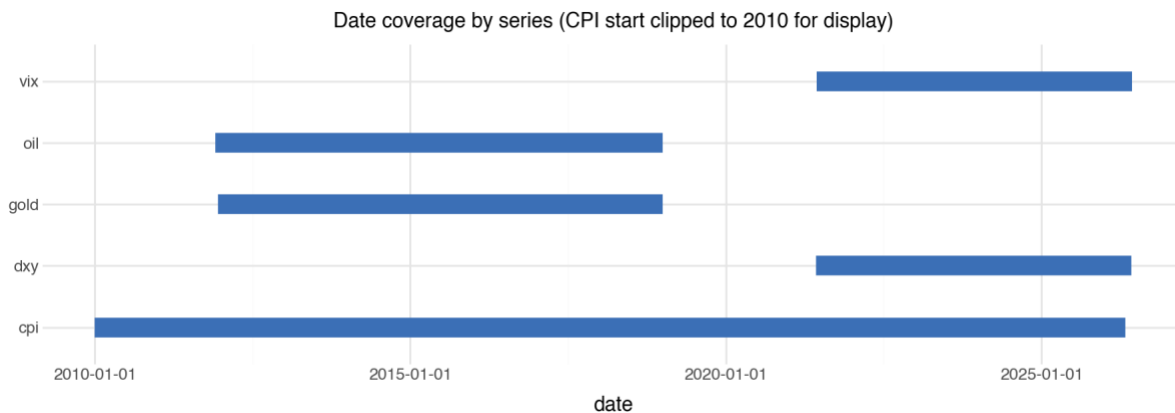


Figure 1. Date coverage by series. The price series and the macro stress series do not overlap.

4. Price behaviour

Across the 85-month window, gold fell about 23 percent and oil about 51 percent. This was a commodity bear market, the decline from the 2011 gold peak and the 2014 to 2016 oil collapse, not a safe-haven period.

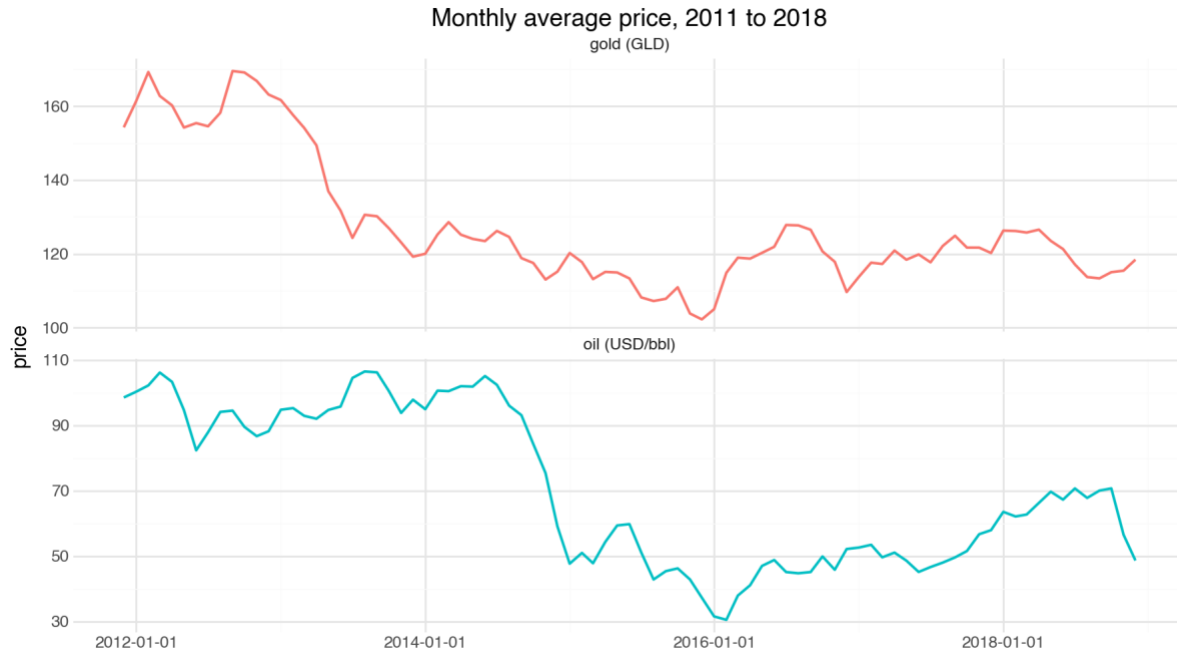


Figure 2. Monthly average price for gold and oil, 2011 to 2018.

5. Volatility

Oil was about twice as volatile as gold from month to month, with mean within-month coefficients of variation near 0.033 for oil and 0.016 for gold. The most volatile oil months line up with known events: the December 2014 price crash, the early 2016 low, and the late 2018 selloff.

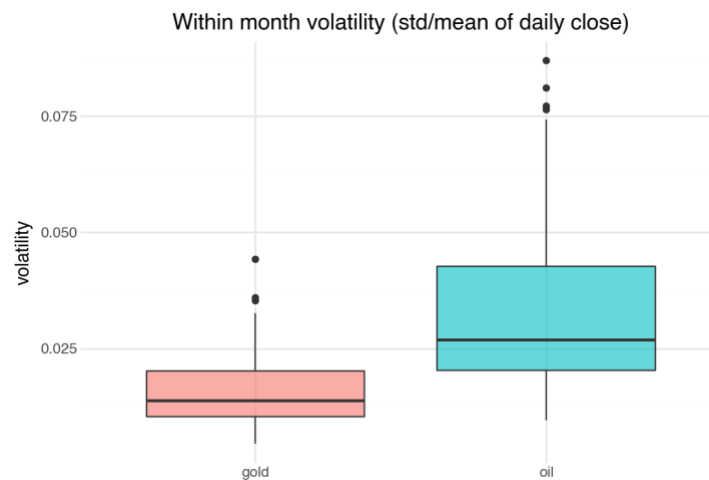


Figure 3. Within-month volatility, gold versus oil.

6. Trading volume

Trading activity is one of the Gold-level questions. Gold volume peaked around February 2016, during the rush into safer assets at the cycle low, while oil volume rose steadily as prices recovered into 2017 and 2018.

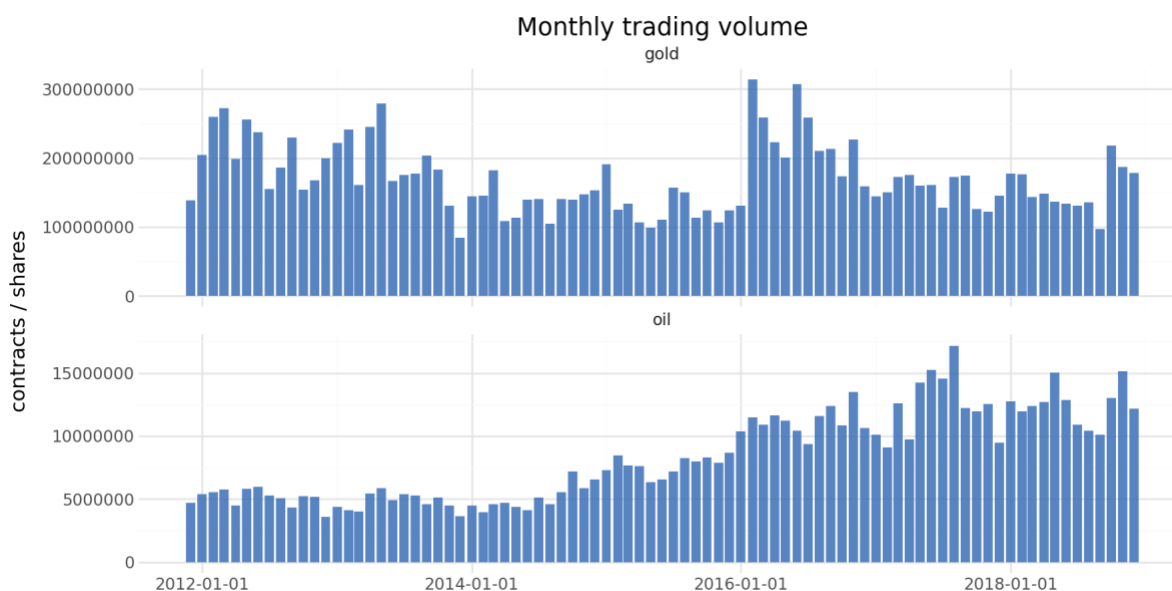


Figure 4. Monthly trading volume for gold and oil.

7. The gold and oil relationship

On price levels gold and oil look strongly linked, but that is mostly a shared downward trend. On monthly returns the link is close to zero, about 0.01, and the rolling correlation moves between strongly negative and strongly positive. The pipeline's rolling correlation, which includes the short early windows, averages about **0.182** and ranges from -0.847 to 1. The notebook's strict twelve-month version averages near 0.12. Either way the message is the same: report this relationship on a rolling basis, because a single number hides it.

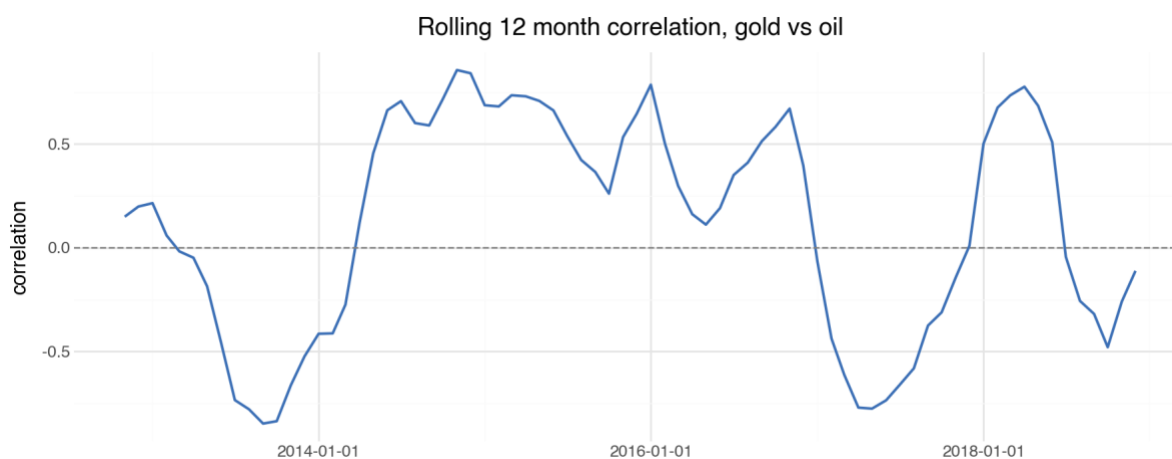


Figure 5. Rolling twelve-month correlation between gold and oil.

8. Hypothesis results

The six hypotheses were run through the same statistics the SQL defines. One test had data and was not significant, one was undefined for lack of variation, and four returned no rows because the data they need does not overlap. Reporting four of six as rejected, which a careless run might do from empty joins, would be false.

ID	Hypothesis	Sample	Result (from SQL run)
H1	Gold higher when the dollar weakens	no data	Untestable: no dollar overlap
H2	Gold higher in high-inflation months	n=45 vs 28	effect -0.49, t -0.60, p 0.55: fail to reject H0
H3	Oil higher when world output falls	n=1 vs 6	Undefined: only one cut year
H4	Gold higher in stressed vs calm months	no data	Untestable: no VIX overlap
H5	Gold returns track changes in VIX	no data	Untestable: no VIX overlap
H6	Gold moves inversely to the dollar	no data	Untestable: no dollar overlap

Table 2. HYPOTHESIS_RESULTS, produced by the executed pipeline using its own p-value macro.

9. Oil supply detail

The oil-supply test (H3) is undefined because world output rose in all but one year of the overlap, leaving a single year in the falling-output group. The annual view below shows why a fair two-group comparison cannot be formed.

Year	Avg oil (USD)	World output (mbd)	Oil return YoY %	Output change
2011	98.58	75622		
2012	94.15	75764	-4.5	142
2013	98.05	75600	4.2	-164
2014	92.82	79072	-5.3	3472
2015	48.88	83436	-47.3	4364
2016	43.41	84702	-11.2	1267
2017	50.91	85225	17.3	523
2018	64.83	85434	27.3	209

Table 3. V_OIL_ANNUAL: annual oil price against world petroleum output.

10. Inflation hedge

Gold did slightly worse in higher-inflation months, and the difference was not significant (p about 0.55). In this window, with rising real interest rates, the inflation hedge was not visible. The relationship should be treated as conditional, not as a rule.

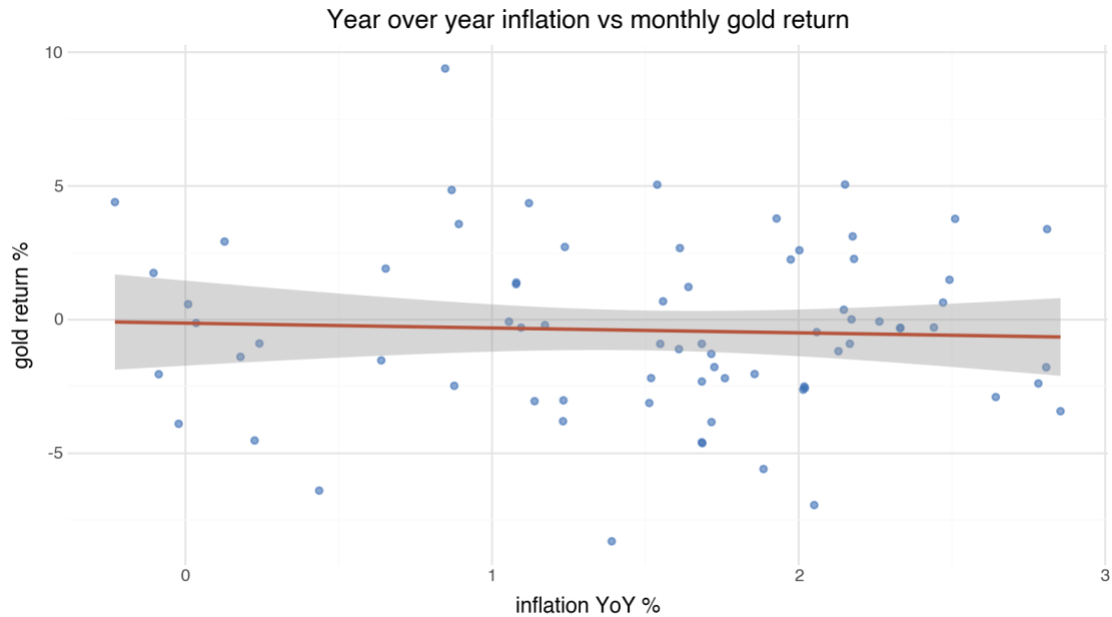


Figure 6. Year-over-year inflation against monthly gold return, with fitted line.

11. Behavioural grouping

As an exploratory step, the months were grouped by behaviour using a scikit-learn pipeline (median imputation, standard scaling, then KMeans), with PCA used to project the groups into two dimensions. The groups separate calm months from high-volatility months. This is a lens for further questions, not a trading signal.

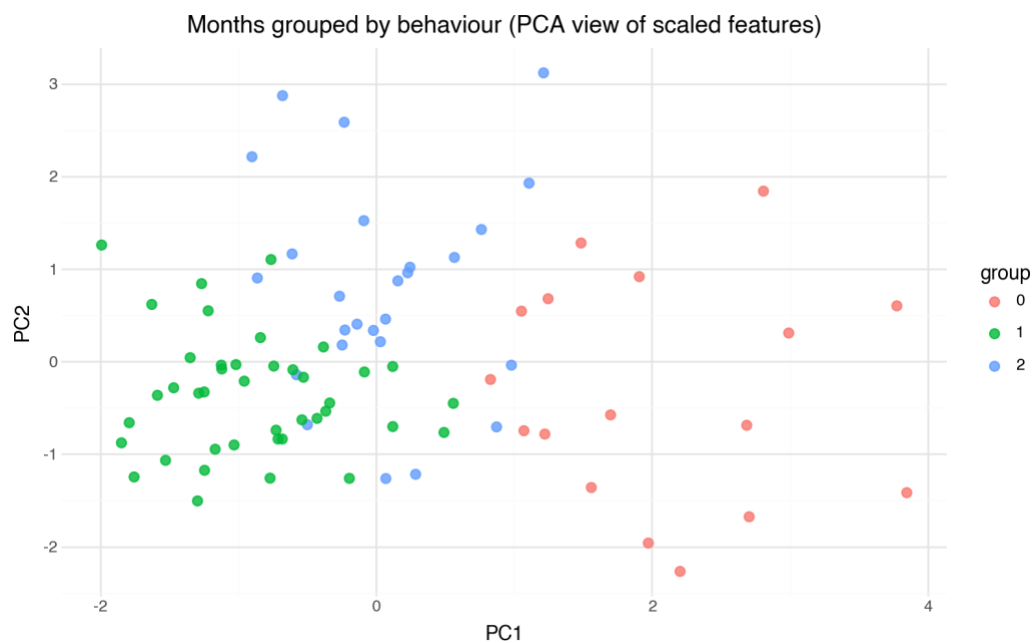


Figure 7. Months grouped by behaviour, PCA view of the scaled features.

12. Analyses that returned no rows

Three further analyses defined in the pipeline produced nothing on this data, and are recorded here rather than hidden. The event-impact study is empty because the events are dated 2022 to 2024, after the prices end. The VIX volatility-regime profile is empty because no price month has a VIX reading. The gold-dollar and gold-VIX rolling correlations are empty for the same reason. All three are sound analyses pointed at data this set does not contain.

13. Recommendations

First, align the data in time: obtain gold and oil for 2021 to 2026, or the dollar and VIX back to 2011, so the cross-asset tests can run. Second, re-align the event study to a period the prices cover. Third, treat the inflation hedge as conditional and re-test once overlapping data exists. Fourth, use world oil supply only as a slow structural backdrop until a true monthly crude series is available. Fifth, always report the gold-oil relationship on a rolling basis.

14. Method note

The pipeline was executed in Snowflake. The cleaning, monthly aggregation, rolling correlation formula, hypothesis tests, and the normal-approximation p-value macro were ported faithfully from the SQL. The small difference between the SQL p-value (0.5496 for H2) and the notebook value (0.5526) reflects the SQL using a normal approximation while the notebook uses the t-distribution. Both lead to the same verdict. Figures were produced by the EDA analysis notebook on the same source files.